

pH Sensitivity of the SERF1a Conformational Ensemble

Shu-Yu Huang, Orion Shih, U-Ser Jeng, Chi-Fon Chang, Jung-Hsin Lin,* and Thérèse E. Malliavin*

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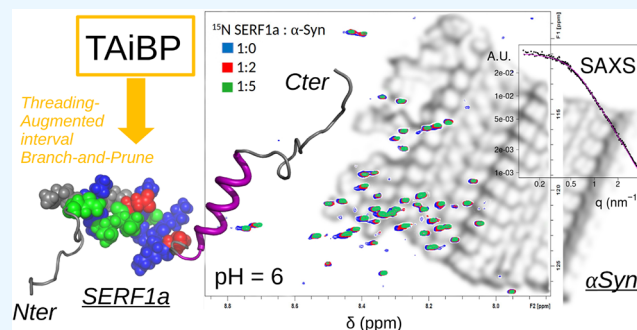


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ABSTRACT: The MOAG-4/SERF class of proteins is a positive regulator of the aggregate formation of amyloid proteins that play an important role in the origin of numerous age-related diseases. In this class of proteins, previous structural studies showed that SERF1a is a very flexible protein, containing at least one α -helical region. The present work proposes the first description of the conformational space of SERF1a at two pH values (6 and 6.8) using nuclear magnetic resonance (NMR) and small-angle X-ray scattering (SAXS). SERF1a NMR conformations have been generated using the CYANA-FLYA procedure as well as the TAiBP (Threading-Augmented interval Branch-and-Prune) procedure, based on the systematic enumeration of protein conformations within an original distance geometry scheme. The generated conformations were filtered using Pepsi-SAXS by fitting small-angle X-ray experiments based on size exclusion chromatography (SEC-SAXS). At pH 6.8, a good fit of the SEC-SAXS curves was obtained for the CYANA and TAiBP NMR conformations, while at pH 6, analysis of the NMR conformations and the SEC-SAXS curves indicated that there could be a conformational exchange between a compact conformation displaying long-range nuclear Overhauser effects (NOEs) between residues LYS-13 and THR-32, and slightly more extended conformations. The shortening of the C-terminal α -helix, as well as the destabilization of the N-terminal α -helix at acidic pH, may be related to the physiological function of SERF1a in the nucleoli. Interestingly, the N-terminal region of SERF1a displays numerous possible binding pockets, particularly in the region interacting with α -synuclein, detected by chemical shift perturbations. In general, the TAiBP procedure allows for a more expanded exploration of the SERF1a conformational space as well as for a description of the internal dynamics of SERF1a in agreement with NMR relaxation and with predictions from the protein primary sequence.



INTRODUCTION

Protein disorder plays a key role in numerous biological processes because of the versatility of interactions that flexible regions can establish with other biomolecules. The Small EDRK-Rich Factor (SERF) class is a particularly striking example of such versatility.^{1,2} Indeed, the physiological, normal function of this protein³ was identified as an RNA-chaperoning activity that favors the incorporation of RNA into nucleoli and liquid-like artificial RNA organelles.^{4,5} However, the N-terminal region (residues 6–28) of SERF1a was shown^{6,7} to interact with pro-amyloid α -synuclein under conditions of cellular stress. In a similar way, the “modifier of aggregation 4” (MOAG-4), as well as SERF2, drives the aggregation and toxicity of amyloidogenic proteins by interacting with their negatively charged segments.^{8,9}

This interaction between SERF1a and amyloid proteins has attracted specific interest, as SERF1a promotes the oligomerization and fibril appearance of α -synuclein,¹⁰ as well as the separation of liquid–liquid phase.^{4,11} In addition, SERF1a shows possible interactions with other amyloid proteins, such as NT17-polyQ peptides of huntingtin.¹² Thus, interfering with the interaction between SERF1a and amyloid proteins may provide a way to impact amyloid fibrillation.¹³

The class of MOAG-4/SERF proteins has been studied in recent years^{2–9,11–14} using strategies that involve structural biology and molecular modeling. This family of partially folded proteins displays one or two α -helical segments.^{1,8} Nuclear Magnetic Resonance (NMR) experiments⁸ have provided a set of MOAG-4 conformations calculated from NMR chemical shifts.

In the present work, an NMR study of the small EDRK-rich factor 1 isoform 1 of *Homo sapiens* (NP_001171558.1) was carried out at two different pH values. Two approaches were used in parallel to characterize the conformational space of SERF1a. First, the pipeline CYANA-FLYA¹⁵ allowed us to propose a set of conformations containing long N- and C-terminal α -helices. In addition, CYANA-FLYA has assigned a long-range NOE at acidic pH, connecting the residues LYS-13

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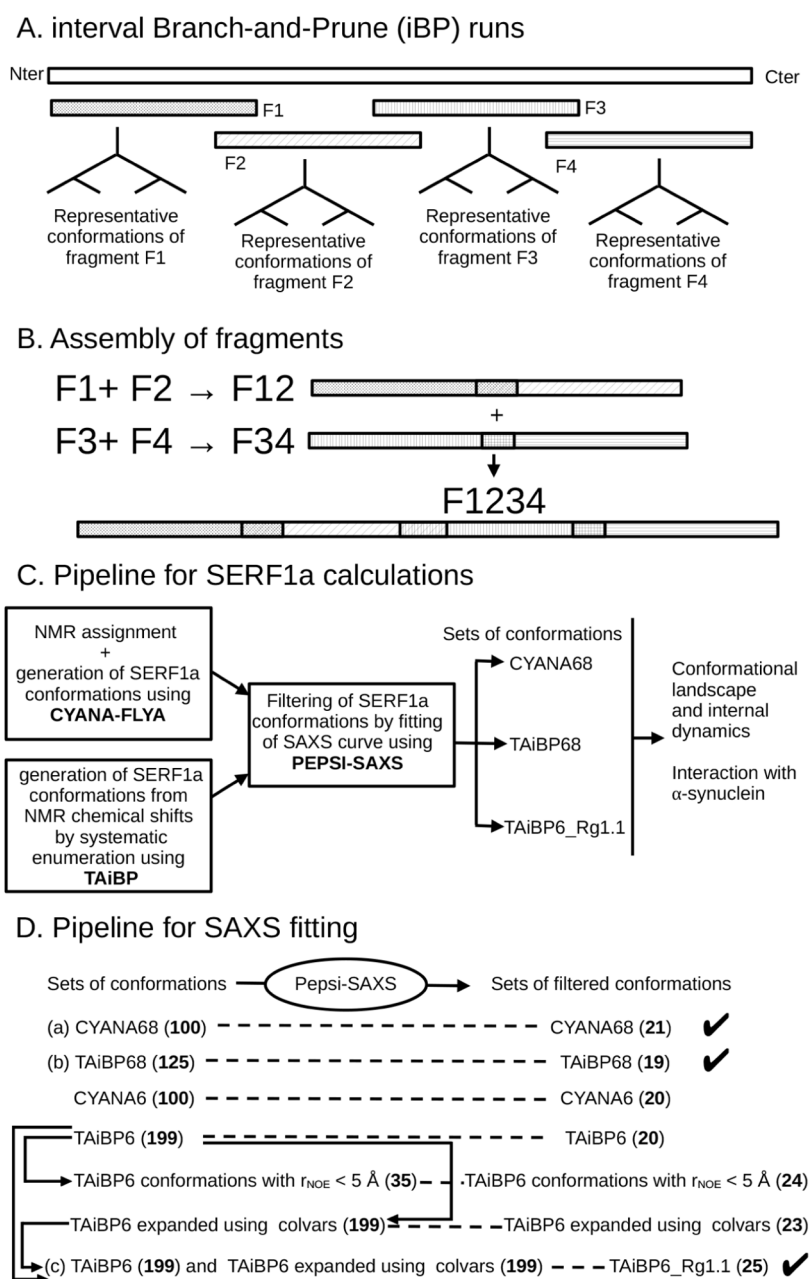


Figure 1. (A, B) Scheme describing the TAI BP procedure. For the sake of clarity, the procedure is described using only four fragments. The two steps are: (A) interval Branch-and-Prune (iBP) runs, enumerating exhaustively the fragments' conformations by building trees and producing representative conformations using the self-organizing map approach.²⁷ (B) assembly of successive two-by-two fragments performed hierarchically to obtain conformations of the backbone of the full protein. The side chains are then added and refined using molecular dynamics simulations. (C) Pipeline of the calculations performed here on the protein SERF1a. (D) Pipeline of the Pepsi-SAXS filtering. The labels a–c correspond to the panel labels in Figure 3. The number of conformations is written in bold in parentheses, and the selected sets of conformations are indicated with a check mark sign.

and THR-32, located in the N- and C-terminal helices, and produced a set of compact conformations.

Second, the approach Threading-Augmented interval Branch-and-Prune (TAiBP), designed for intrinsically disordered proteins (IDPs), was used to generate SERF1a conformations. Indeed, it is difficult to characterize the conformational landscape of IDPs, as they sample highly variable conformations of similar energies. Thus, the structural convergence criterion used in classical NMR structure determination does not apply here. Therefore, the analysis of IDP conformational spaces classically relies on the use of

Monte Carlo approaches, used in particular in Flexible-Meccano^{16,17} and molecular dynamics simulations,¹⁸ as, for example, in the ENSEMBLE approach.¹⁹ The purely geometric TAI BP approach was also proposed^{20–22} to systematically enumerate the conformations of intrinsically disordered proteins and to determine a set of representative conformations. The inputs of the TAI BP calculations were intervals of torsion angles ϕ and ψ , obtained from the definition of α -helices and from δ 2D predictions²³ for the other protein regions (Figure 1A,B).

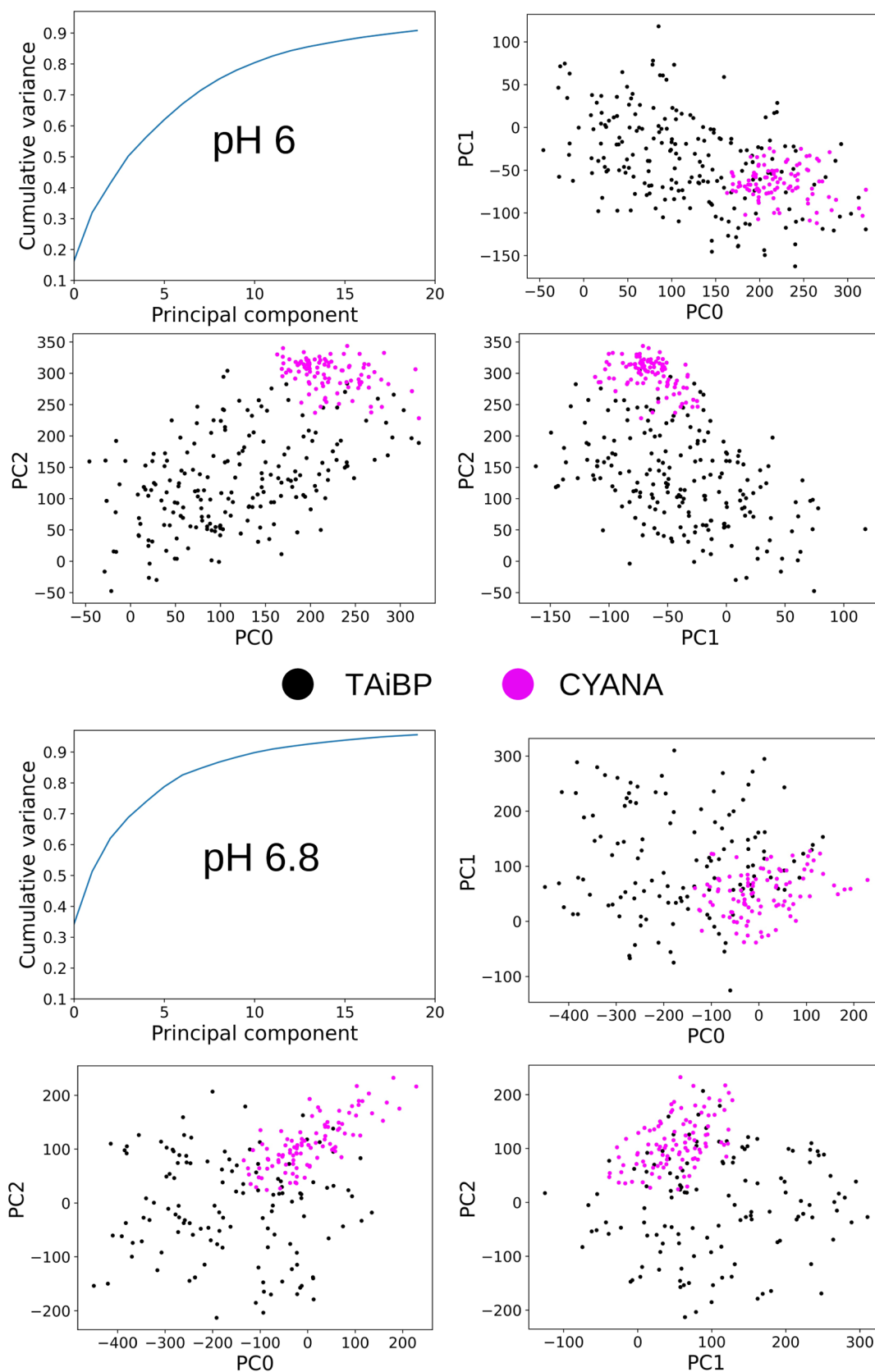


Figure 2. Principal component analysis (PCA) of the SERF1a CYANA and TAIiBP NMR conformations obtained at pH values of 6 (four top panels) and 6.8 (four bottom panels). For each set of four panels, the upper-left panel describes the variation of the cumulative sum of eigenvalues, sorted in decreasing order. The three other panels describe the projection of the SERF1a conformations on the planes defined by eigenvectors corresponding to the first-second, second-third, and first-third eigenvalues, sorted in decreasing order. The projected CYANA and TAIiBP conformations of SERF1a are colored magenta and black.

The CYANA and TAIiBP NMR conformations were then filtered by fitting SEC-SAXS curves using the Pepsi-SAXS approach²⁴ (Figure 1C) and producing three sets of conformations: CYANA68, TAIiBP68, and TAIiBP6_Rg1.1 (Figure 1D). The Pepsi-SAXS approach has the advantage of providing a fairly accurate description of the hydration shell.²⁴ The set of conformations calculated by using CYANA at pH 6 was not selected by filtering with the SEC-SAXS curves. In fact, the long-range NOE extracted by CYANA-FLYA at pH 6 induced conformations that did not fit the SEC-SAXS curve.

The current study differs greatly from prior SERF or MOAG-4 structural studies, mainly due to the use of the TAIiBP approach. Indeed, the MOAG-4 conformations were obtained from molecular dynamics trajectories and by applying restraints from the chemical shifts.⁸ In other previous studies of SERF proteins, molecular modeling calculations were performed. A set of SERF2 conformations was obtained from *ab initio* modeling and submitted to molecular dynamics simulations.²⁵ All-atom Monte Carlo simulations were performed on *S. cerevisiae* SERF⁴ using the CAMPARI/ABSINTH package.²⁶ But, no representative sets of conformations have been extracted from these calculations. In addition, the previous structural studies are not based on the δ 2D predictions. The TAIiBP approach described here has the specificity to permit a systematic enumeration of possible conformations according to the δ 2D predictions.

The SAXS-filtered conformations were then compared to the experimental observations and predictions on internal dynamics, allowing to reconcile the disordered and non-disordered descriptions of this protein (Figure 1C). The analysis of putative binding pockets reveals conformations prone to intermolecular interactions, especially in the N-terminal region, in agreement with previous experimental observations⁶ as well as with the titration performed here between SERF1a and α -synuclein.

RESULTS

Observation of Long-Range NOEs at Acidic pH Produces a Rather Compact Set of CYANA Conformations for SERF1a

Among the sets of conformations CYANA6 and CYANA68 obtained at the two pH values using CYANA-FLYA,¹⁵ the root-mean-square deviation (RMSD, Å) between the atomic coordinates reveals a fairly flexible set of conformations (Figure S1), with all protein residues located in the most favored and additionally allowed Ramachandran regions and no restraint violations (Table S1). Consensus α -helices are observed for residues 6–14 and 32–53 at pH 6, and for residues 6–14 and 32–52 at pH 6.8. A good definition of each α -helix was attained, as pairwise r.m.s. deviations between backbone atomic coordinates are in the range 0.1–1.3 Å within each α -helix (Table S1). The range of residues corresponding to consensus α -helices, located in the N- and C-terminal moieties of SERF1a, will be denoted in the following as hN and hC, regardless of whether secondary structure α is formed or not.

These observations on SERF1a are similar to the observations made previously on MOAG-4⁸ and, more recently, on human SERF2.^{4,5,14} Furthermore, the shorter length of hN α -helix is in qualitative agreement with the presence of a low-populated N-terminal α -helix in MOAG-4 at pH 6.⁸

More compact conformations are observed at pH 6, displaying a smaller root-mean-square deviation (RMSD, Å)

of atomic coordinates (Table S1) on residues 6–14 and 32–52. This is induced by the presence of two long-range nuclear Overhauser effects (NOEs) between the amide hydrogen of THR-32 and the methylene groups C γ and C δ of LYS-13, with target distances of 2.39 and 2.55 Å for the corresponding restraints.

The TAIiBP Approach Using δ 2D Predictions Provides a Wider Sampling of the Conformational Space

TAiBP68 and TAIiBP6 runs have been performed on SERF1a, using the ϕ/ψ boxes extracted from the δ 2D prediction²³ at the two pH values. The TAIiBP procedure, described in more detail in the Supporting Information, is summarized here. At each pH value, interval values for dihedral angles ϕ and ψ were determined from the δ 2D prediction²³ using statistics on Ramachandran regions. This procedure is described in more detail in the section "Determination of ϕ/ψ input boxes from NMR chemical shifts" of the Supporting Information. The protein SERF1a was divided into a set of 11 fragments (Table S2) and the possible conformations of each fragment were enumerated using the interval Branch-and-Prune (iBP) approach (Figure 1A). To avoid an exponential increase in the number of possible conformations, we systematically clustered them to extract representative conformations. Once a set of representative conformations was determined for each fragment, successive fragments in the protein sequence were assembled two by two by superimposing the coordinates of their extremities (Figure 1B). These assembly steps, followed by clustering steps, were realized in a hierarchical way²⁸ to obtain a set of representative conformations of the backbone of the whole protein. The protein side chains were then added using a molecular dynamics procedure described in the section "Addition of side chains to TAIiBP conformations" of the Supporting Information. At the end of the TAIiBP procedure, 125 and 199 conformations were finally obtained for the TAIiBP68 and TAIiBP6 runs.

At each pH value, the conformational space spanned by the CYANA and TAIiBP conformations was compared using Principal Component Analysis (PCA) implemented in the MDAnalysis Python library.²⁹ The TAIiBP conformations were pooled with the CYANA conformations, and the full set of atomic coordinates was analyzed by PCA, superimposing all conformations by minimizing the root-mean-square of the coordinate differences. The first 20 PCA eigenvalues provide a description of more than 90% of the variation in the atomic coordinates (Figure 2). Then, each SERF1a conformation was projected onto the planes defined by the eigenvectors corresponding to the three largest eigenvalues (Figure 2). The projections of CYANA conformations, colored magenta, are less scattered than the whole set of projections, showing that the use of TAIiBP has expanded the exploration of conformational space with respect to CYANA.

Filtering by SEC-SAXS Data Reveals Underlying Conformational Variability of SERF1a at Acidic pH

The superimposed SEC-SAXS data measured at the two pH values display discernible differences (Figure S2). The Guinier analyses of these curves, performed using the command AUTORG in ATSAS-3.2.1–1³⁰ give also significantly different radii of gyration (Figure S3a). At pH 6.8, 60 data points (according to the criterion $qR_g < 1.0$, corresponding to the Guinier approximation) were used in the analysis and produced a radius of gyration of 26.6 ± 0.8 Å with a quality of 98%. Using the same criterion, at pH 6, 62 data points were

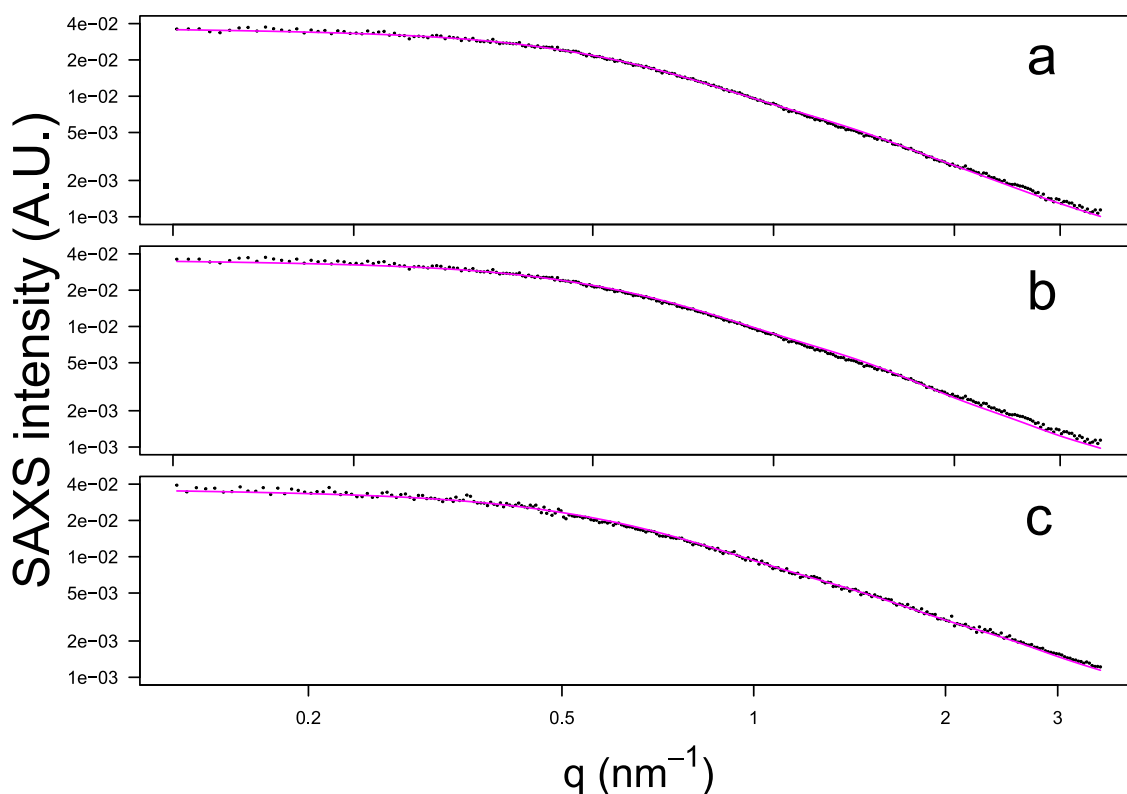


Figure 3. Fitting of the SEC-SAXS curves measured at pH 6.8 (a, b) and at pH 6 (c) using Pepsi-SAXS.²⁴ The SEC-SAXS curves calculated from the selected conformations (see the section “Filtering by SEC-SAXS data reveals underlying conformational variability of SERF1a at acidic pH” of the main text for the details of the selection method) are plotted as magenta lines along with the SAXS intensities measured experimentally (black points). The axes x and y are on a logarithmic scale. The fits are obtained by starting from the following conformation sets: CYANA68 (a) and TAI6P68 (b), obtained from NMR data measured at pH 6.8 and (c) the TAI6P6 conformations pooled with the TAI6P6 extended conformations, for which the radius of gyration has been scaled by 1.1 using a colvars restraint,^{32,33} applied along a molecular dynamics trajectory.

Table 1. Results of the Pepsi-SAXS Fitting^a

Set of conformations	Pepsi-SAXS fitting	Number of selected conformations	Threshold χ value	χ^2 value	Radius of gyration (nm)
*CYANA68	3a	21	1.36	0.11	2.43 ± 0.18
*TAI6P68	3b	19	1.55	0.21	2.31 ± 0.16
CYANA6	-	20	1.75	0.77	1.90 ± 0.10
TAI6P6	-	20	1.26	0.37	2.08 ± 0.11
TAI6P6 conformations with $r_{\text{NOE}} < 5 \text{ \AA}$	-	24	2.14	1.11	1.83 ± 0.12
TAI6P6 conformations expanded using colvars ^{32,33}	-	23	0.93	0.18	2.26 ± 0.12
*TAI6P6_Rg1.1 conformations pooled from expanded TAI6P6 and TAI6P6 conformations	3c	25	0.92	0.18	2.26 ± 0.11

^aThe label of the panel displaying the fit (Figures 1D and 3) is given, as well as the number of selected conformations, the threshold χ value, the $\chi^2 = \frac{1}{N-1} \sum_{j=1}^N \left[\frac{I_{\text{exp}}(q_j) - I_{\text{calc}}(q_j)}{\sigma(q_j)} \right]^2$ value and the average radius of gyration. The conformation sets kept for further analysis are marked with an asterisk. The colvars procedure is described in the main text.

used in the analysis and produced a radius of gyration of $24.4 \pm 1.0 \text{ \AA}$ with a quality of 97%. The Kratky plots generated from the SAXS curves display shapes characteristic of noncompact partially unfolded proteins (Figure S3b). The corresponding pair distance distribution functions $P(r)$ (Figure S3c) were obtained using the command GNOM of ATSAS 3.2.1³⁰ and display very small differences. Although the analysis of SAXS curves displays known limitations for intrinsically disordered proteins,³¹ the availability of CYANA and TAI6P conformations made it possible to obtain a good fit of the two sets of SAXS curves, as shown below.

The software Pepsi-SAXS was used to calculate the simulated SAXS curves from the TAI6P and CYANA conformations. For a given set of CYANA or TAI6P conformations, the fit of each conformation to the SEC-SAXS curve was evaluated using the reduced parameter $\chi = \sqrt{\chi^2}$ calculated by Pepsi-SAXS from the simulated and experimental curves. The analyzed conformations were sorted according to the obtained $\chi = \sqrt{\chi^2}$ values. A threshold value for χ is then determined in order to select around 20 conformations corresponding to plateaus of χ values (Figure S4, left column). A smaller subset of conformations with low χ

values is defined from the ensemble of a whole range of χ values.

The Pepsi-SAXS analysis performed on the sets of conformations CYANA68, TAI6P68, CYANA6, and TAI6P6 reveals that the best fits are obtained for CYANA68 (Figure 3a) and TAI6P68 (Figure 3b) with χ^2 values of 0.11 and 0.21 (Table 1). On the contrary, for CYANA6 and TAI6P6, larger χ^2 values of 0.77 and 0.37 are observed (Table 1). In order to investigate the reasons for which a worse fit was obtained for TAI6P6, several biasing protocols were applied to the set of initial TAI6P6 conformations presented to Pepsi-SAXS for filtering. The selection of TAI6P6 conformations for which the ambiguous distance r_{NOE} between the hydrogens of LYS-13 and THR-32 was less than 5 Å produces a more deficient fit, with a χ^2 value of 1.11 (Table 1) even greater than those of the previous fits on CYANA6 and TAI6P6.

Interestingly, the deterioration of SAXS fitting for TAI6P6 in the presence of the long-range NOE corresponds to a decrease of the averaged radius of gyration down to 1.8 nm (Table 1). Thus, it was investigated whether starting from TAI6P6 conformations and imposing an increase in the radius of gyration would allow a better fit of the SEC-SAXS curve. For that purpose, the TAI6P6 conformations have been extended using a colvars restraint^{32,33} applied on the radius of gyration with a target value that scales the radius by 1.1. The expansion was carried out over 50,000 steps of molecular dynamics trajectory starting from each TAI6P6 conformation and using a force constant of 10 kcal/mol/Å².

The use of these extended conformations significantly improved the fit, allowing one to reach a χ^2 value of 0.18 (Table 1). Then, the 199 initial TAI6P6 conformations were concatenated with the 199 extended TAI6P6 conformations, and the whole set of 398 conformations was then filtered by Pepsi-SAXS, producing the same χ^2 of 0.18 and a good fit of the SAXS curve (Figure 3c). In the following, the set of conformations filtered from this pooled set of conformations, named TAI6P6_Rg1.1 (Table 1), will be used as the representation of the SERF1a conformational space at pH 6, as it describes a variety of protein conformations in agreement with the observation of the long-range NOE, while allowing the explanation of the SEC-SAXS curve. Indeed, in the set TAI6P6_Rg1.1, 2 initial TAI6P6 conformations and 23 TAI6P6 extended conformations were selected.

The various fits of SAXS curves (Figures 3 and S5) show that the existence of a stable long-range NOE contradicts the value of the radius of gyration observed by SAXS. The procedure to obtain a set of conformations TAI6P6_Rg1.1 was able to reconcile these contradicting pieces of information. The three error-weighted residual difference plots³⁴ all display small deviations of the simulated SAXS profiles from the data, supporting the model fitting quality³⁵ (Figure S6). Nevertheless, it should be noted that the procedure followed to produce the conformations TAI6P6_Rg1.1 is greatly facilitated by the presence of a stable α -helix structure for residues 33–47, due to the observed negative Chemical Shift Index,³⁶ as described in the section "TAI6P6 calculation" of the Supporting Information.

The Conformational Landscape is Dominated by the Variations of the Radius of Gyration and the Angle between hN and hC Regions

Following the filtering of conformations using the SEC-SAXS data, three sets of conformers obtained at pH 6.8 (CYANA68,

TAI6P68) and at pH 6 (TAI6P6_Rg1.1) have been selected as a reliable description of the conformational space of SERF1a. The conformation superimposition is quite different for the CYANA68 or TAI6P6 conformations (Figure 4), due to the

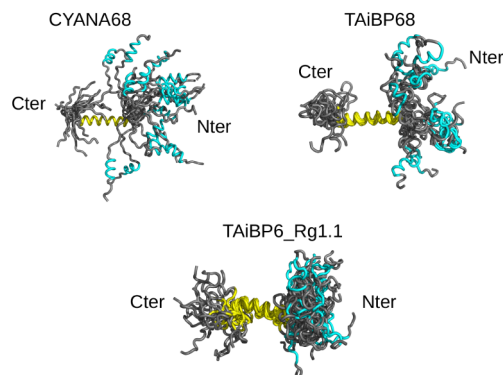


Figure 4. Sets of CYANA and TAI6P6 conformations, filtered by SEC-SAXS curve fitting, are superimposed on the α -helix hC which is colored in yellow. The hN region is colored in cyan. The hN and hC regions correspond to the consensus α -helices detected by NMR and located in the N- and C-terminal moieties of the protein. The sets of conformations correspond to those marked with an asterisk in Table 1.

well-determined α helix for hN in the CYANA68 conformations. In contrast, in TAI6P68 conformations, the hN region displays not only α -helix but also 3–10 helix and turn structures (Figure S7), producing sets of more diverse conformations.

The region between hN and hC also shows different features between the CYANA and TAI6P6 conformations. For TAI6P68, this region is equally populated as coil and turn, while for CYANA68, it shows more coil than turn (Figure S7), in agreement with the more extended conformations observed for this subset.

These observations can be connected to the chemical shifts measured on proteins belonging to the SERF family. Three sets of chemical shifts—bmr51051, bmr51829, and bmr52619—recorded on SERF proteins and deposited in BMRB³⁷ were analyzed using TALOS-N³⁸ to detect the regions that form α helices. For the protein SERF1a, sets bmr51051 and bmr52619 were recorded at pH 6.5, while the set bmr51829 was recorded at pH 6.8. At the more acidic pH (6.5), α helices were detected only in the C-terminal part (bmr51051: residues 35–63, bmr52619: residues 36–37), while longer α -helices were detected in the N- and C-terminal parts at a higher pH value (6.8) (bmr51829: residues 8–12, 34–47, 51–52). For human SERF2 (bmr52141 data set), TALOS-N predicted three very short helices (3–4 residues long) in the N-terminal region of the protein and a long helix in the C-terminal region spanning residues 37–46.⁵ These predictions of α helices agree with the helices detected here in the filtered conformations and with the effect of acidic pH: the disappearance of N-terminal α helices and the shortening of C-terminal α helices.

A simplified description of the conformational space of SERF1a is obtained by plotting the radius of gyration (Å) along with the angle between the local axes along the residues of the hN and hC regions (Figure S8). This description reveals similar features characterized by a slight anticorrelation between these two parameters. More extended conformations

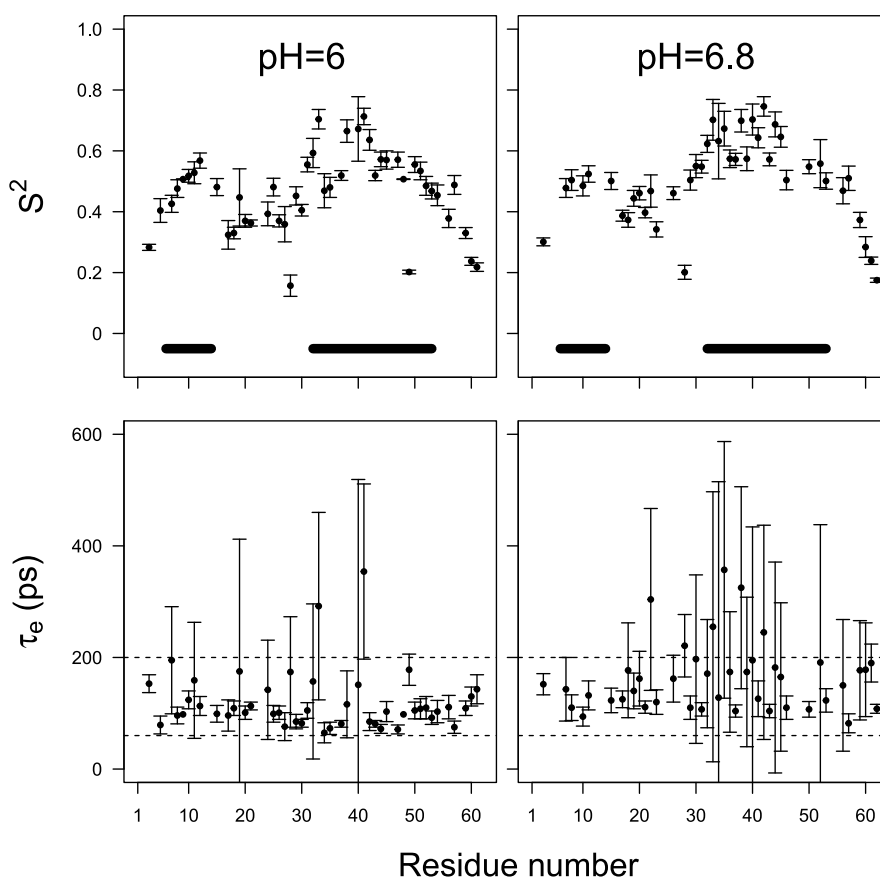


Figure 5. Internal dynamics analysis of ^{15}N – ^1H relaxation was performed using the model m2 of relax.⁴¹ The order parameters S^2 and the local correlation time (τ_e , ps) are plotted along with their error bars. The corresponding values are given in Tables S5 and S6. The positions of the α -helix consensus regions hN and hC are indicated with thick black lines at the bottom of the S^2 panels. The dashed lines are drawn on plots of the local correlation times at τ_e values of 60 and 200 ps.

(with a radius of gyration up to 28 Å) are observed at pH 6.8 (Figure S8, upper panel) than at pH 6 (Figure S8, central panel). In contrast, the set of 946 MOAG-4 conformations, calculated by Yoshimura et al.,⁸ displays more systematic and uncorrelated variations of the radius of gyration and angle. For MOAG-4, the values of the radius of gyration change, of course, toward larger values, due to the larger number of residues in this peptide (82 vs 62 for SERF1a). The anticorrelation between the radius of gyration and the angle is a sign of a relatively correlated set of internal movements in SERF1a. This supports the use of the m2 model to analyze NMR relaxation parameters.

To summarize, the use of combined approaches for generating structures and for filtering conformations permits us to obtain a global picture of the conformational space of SERF1a, characterized by the complementary views provided by CYANA and TAiBP. In fact, at first sight, CYANA produced very contrasting results between pH values of 6.8 and 6, in particular due to the presence of a long-range NOE at pH 6. However, the use of TAiBP, followed by filtering using SAXS data, allowed obtaining much more similar conformational spaces at both pH values (Figure S8).

Measurements of Internal Dynamics Agree with the TAiBP Conformational Space

Comparable profiles of ^{15}N – ^1H NOEs and ^{15}N relaxation rates (Tables S3 and S4 and Figure S9) are observed at both pH values, demonstrating similar internal dynamics of the protein.

The ^{15}N – ^1H NOEs (Figure S9, top panel) display numerous negative values, pointing toward a highly flexible peptide.^{39,40} Interestingly, the region of mostly positive NOE values is observed around residues 31–46, corresponding to the hC region. The profiles of ^{15}N – ^1H NOEs are similar to the ones recently observed for SERF2.⁴

Since a single global correlation time cannot describe intrinsically disordered proteins, not all models for spectral density functions designed for globular proteins are applicable to disordered proteins.⁴² The procedure used here was inspired by the work of Gill et al.⁴³ on the bZip region of GCN4, which is a partially folded protein and consequently shares some structural similarity with SERF1a, which also contains a folded α -helical region.

The analysis of the ^{15}N T1 and T2 relaxation times, and of the heteronuclear ^{15}N – ^1H NOEs, enables us to describe the internal motion of the protein within the framework of phenomenological models. These models⁴⁴ are based on order parameters S^2 , quantifying in the range of 0–1 the degree of mobility and orientation of the ^{15}N – ^1H bond vector within the protein's internal frame of reference, and on the local correlation time of internal motions in SERF1a. Fitting of relaxation data was performed with relax 5.0.0⁴¹ using the global correlation times τ_m of 2.9 ns (pH 6.8) and 2.7 ns (pH 6), estimated by averaging the R2/R1 ratio on residues 32–53 (Figure S10). For each protein residue, among the models fitted with relax, the model giving the smallest value for the Bayesian Information Criterion (BIC)⁴⁴ was selected for

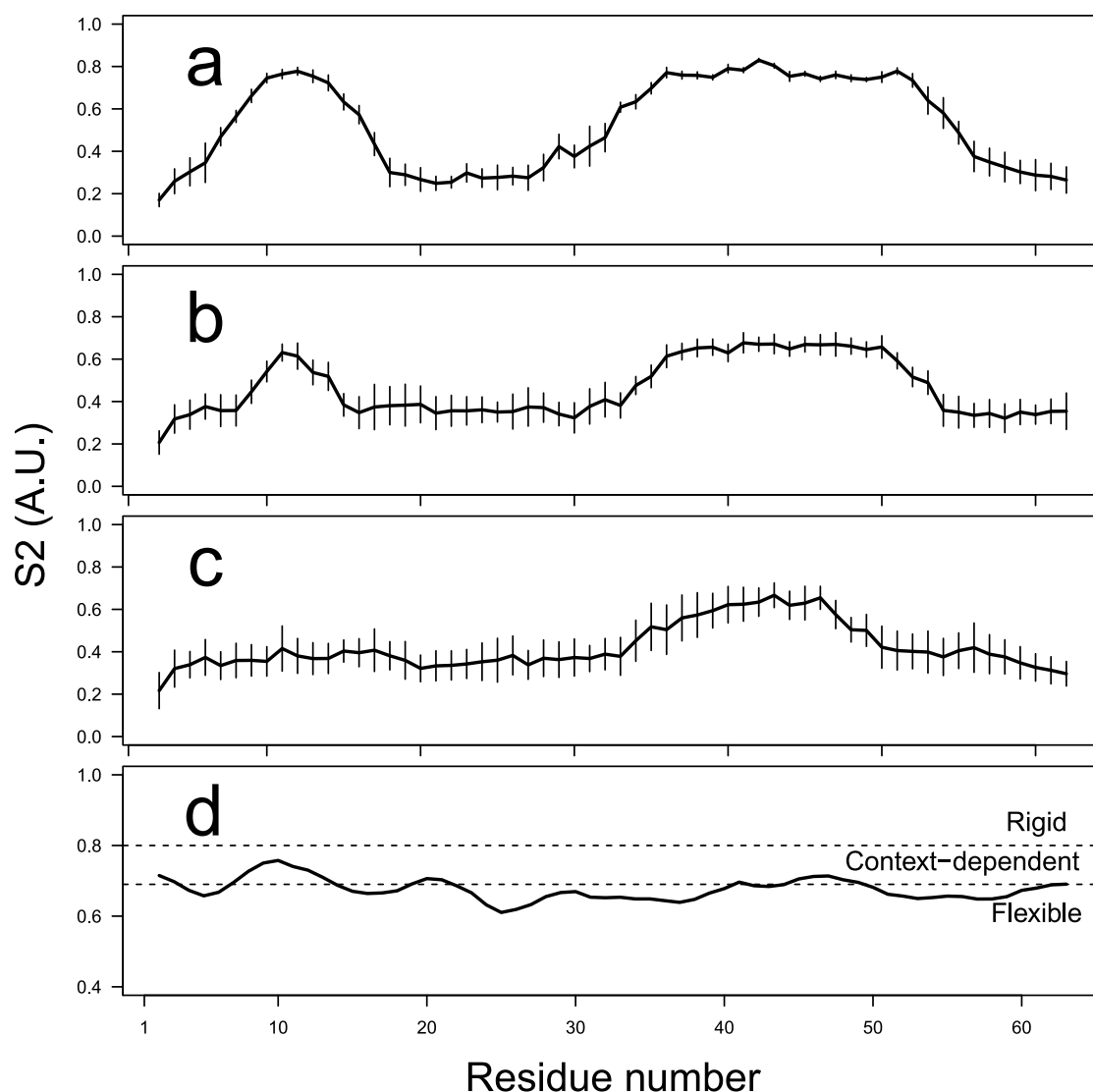


Figure 6. (a–c) Prediction of order parameters S^2 using the model of Zhang and Brüschweiler⁴⁵ on the three sets of filtered conformations: CYANA68 (a), TAI BP68 (b), and TAI BP6_Rg1.1 (c). The displayed S^2 and standard deviation values correspond to the average values calculated over the sets of filtered conformations and are plotted along the SERF1a residue numbers. (d) Prediction of S^2 values using DYNAMINE,^{46,47} from the primary sequence.

further analysis. At both pH values, for most of the analyzed residues (43 out of 47 at pH 6.8 and 48 out of 53 at pH 6), the model m2 was selected, characterized by one order parameter S^2 and an internal correlation time τ_e (Tables S5 and S6 and eq 1 of Supporting Information). This model is now analyzed to provide a more homogeneous description of the internal dynamics. Additional details of the NMR relaxation analysis are given in the Supporting Information (section "Analysis of the relaxation parameters").

The parameters S^2 (Figure 5) are globally similar between the two pH values, in agreement with the previous observation on relaxation parameters (Figure S9). The order parameters S^2 in the range of 0.2–0.7 correspond to a very flexible protein. A local maximum of S^2 values is observed in region hC, corresponding to the most stable α -helix. Another, less visible, local maximum of S^2 values is observed in the region hN. The residues at the extremities of the α -helices display smaller order parameters, in agreement with the variable secondary structures in TAI BP conformations (Figure S7). The internal correlation times τ_e display more variable profiles than the

order parameters S^2 and are mostly located in the 60–200 ps range. Internal correlation times τ_e display larger errors at pH 6.8 than at pH 6, mostly located in the region hC.

Using the equation proposed by Zhang and Brüschweiler,⁴⁵ theoretical S^2 values calculated for each conformation of CYANA68, TAI BP68, and TAI BP6_Rg1.1, and then averaged, are plotted in Figure 6a–c. The patterns of calculated S^2 profiles along the protein sequence are quite similar to the corresponding experimental profiles of S^2 parameters, especially in the region hC and to a lesser extent in the region hN (Figure 5). Since SERF1a is a partially disordered protein, the good agreement between the predictions of the Zhang and Brüschweiler model⁴⁵ and experimental measurements seems surprising. This relationship highlights the special characteristics of SERF1a conformational space, in agreement with the previously observed anticorrelation linking the gyration radius and the angle between the axes of the hN and hC regions (Figure S8).

The method DYNAMINE,^{46,47} was also used to predict the order values of SERF1a from its primary sequence. The

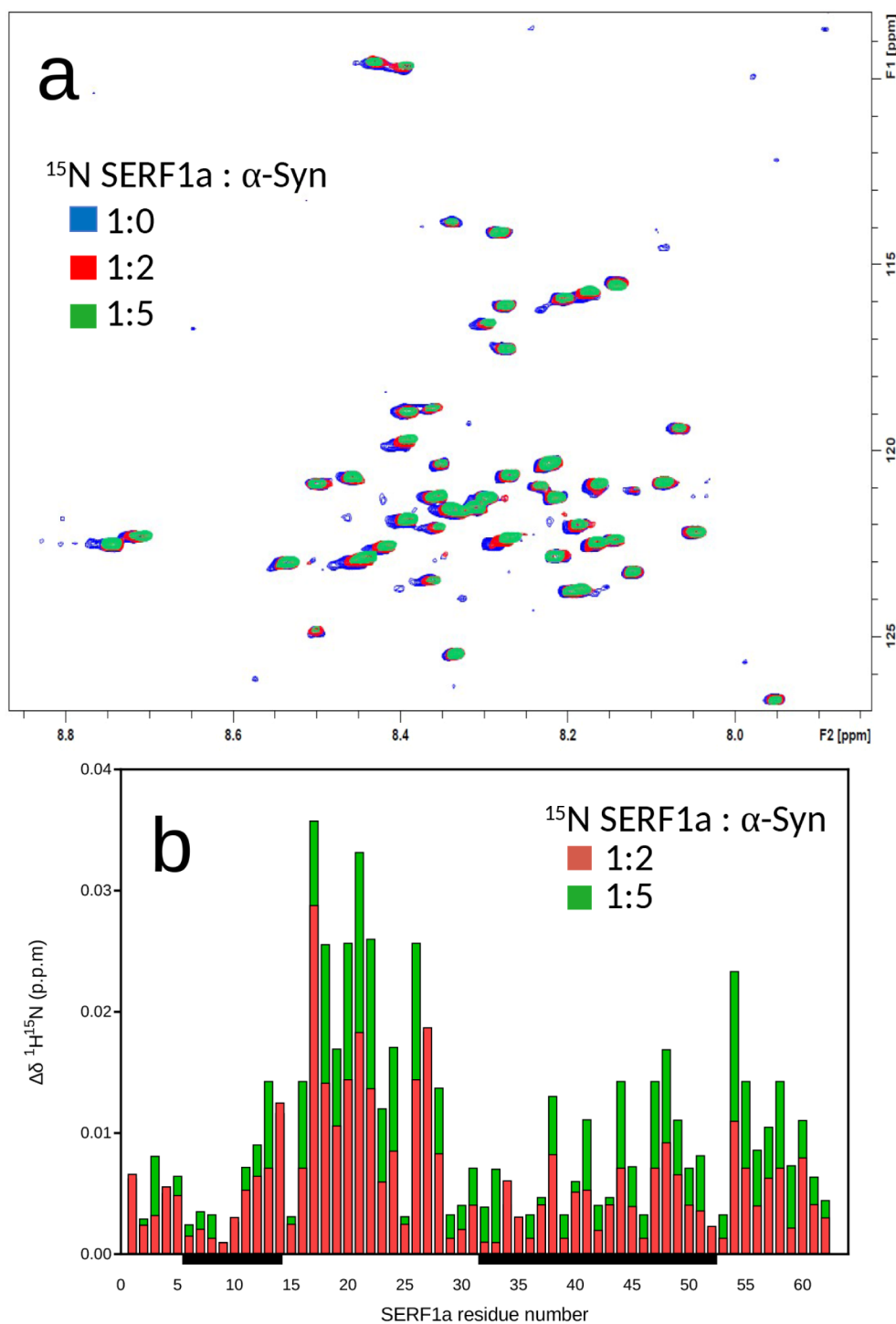


Figure 7. Titration of SERF1a by α -synuclein using 2D NMR ^1H – ^{15}N HSQC spectra. The proteins were dissolved in a buffer containing 20 mM NaPi, 20 mM NaCl, 10% D_2O , at pH 6. (a) Contour plots of 2D NMR ^1H – ^{15}N HSQC spectra using molar ratios 1:0 (blue peaks), 1:2 (red peaks), and 1:5 (green peaks) for SERF1a: α -synuclein. (b) Amide chemical shift perturbations were measured and normalized using the factor: $\delta_{\text{avg}} = \sqrt{(\Delta\delta_{\text{H}})^2 + (0.154\Delta\delta_{\text{N}})^2}$ for the 1:2 (red bars) and 1:5 (green bars) molar ratios. The positions of α -helix consensus regions hN and hC are indicated with thick black lines at the bottom of panel b.

obtained profile (bottom panel in Figure 6d) is in qualitative agreement with the relaxation analysis (Figure 5) as well as with the Zhang prediction at pH 6.8⁴⁵ (Figure 6b) with the presence of bumps in the hN and hC regions. The prediction of these bumps is noted to be context-dependent, which agrees with the differences observed from the Zhang prediction when the pH value varies. The use of completely different

approaches thus produces very consistent results. This supports the overall consistency between the obtained conformations and the relaxation and dynamics analysis, as well as their relation to the intrinsic properties of the SERF1a primary sequence.

Loop Region Connecting hN and hC is a Good Candidate for Interacting Interface with α -Synuclein

Variations in SERF1a conformations in the presence of α -synuclein were investigated by recording 2D ^1H – ^{15}N HSQC spectra at pH 6 using various molar ratios from 1:0 up to 1:5 (Figure 7a). Up to the ratio 1:2 of SERF1a1: α -synuclein, the ^1H – ^{15}N correlation peaks of SERF1a showed limited chemical shift perturbations (CSPs). This behavior agrees with the behavior observed for SERF2 interaction with TAR RNA, in which both partners did not undergo major structural rearrangements.⁴ In contrast, the SERF1a 13–28 residues show increased CSPs for the titration performed with the SERF1a/ α -synuclein 1:5 molar ratio (Figure 7b). Thus, this region could be the binding interaction zone between the two proteins. Isothermal titration calorimetry (ITC) was performed to quantitatively determine the thermodynamic parameters of the interaction between SERF1a and α -Syn at 298 K and pH 6 (Figure S11). An affinity of about 0.8 μM has been determined, which supports the specificity of the interaction, as described in the section "Isothermal Titration Calorimetry Measurements" of the Supporting Information.

Possible binding pockets of SERF1a were detected in the set of SERF1a conformations using the fpocket software⁴⁸ with a minimum and maximum radius of α -spheres equal to 3 and 6 Å, respectively. Pockets displaying a druggability score greater than 0.5 were selected. The residues forming each detected druggable pocket are listed in Table S7, along with the conformation index in the three sets of SAXS-filtered conformations. The number of conformations displaying druggable pockets is 2 for CYANA68, 9 for TAiBP68, and 7 for TAiBP6_Rg1.1, which correspond at most to one-third of the generated conformations. Statistical analysis of the druggable pockets (Figure S12) shows that the solvent-accessible surfaces of the pockets are mostly half polar and half apolar.

For each protein residue, the average number of selected pockets containing this residue was plotted along the protein sequence, producing quite variable profiles (Figure 8). At both pH values, large numbers of druggable pockets are detected in the N-terminal region of TAiBP conformations, particularly in the region of residues 13–28, shown to interact with α -synuclein. The description of the SERF1a conformational space, with a very dynamic N-terminal region that displays exchange between the α and the 3–10 helix (Figure S7) thus allows for an increase in the occurrence of druggable binding pockets and consequently the possible number of interactions with potential partners. This characteristic also agrees with the modest relative compactness observed for the protein SERF2 when bound to RNA.⁴ Examining the positions of acidic/basic residues (Figure 8), the N-terminal residues K13, K16, and K17, for which mutations to E were shown to attenuate the interaction of SERF1a with α -synuclein,⁶ correspond to large numbers of druggable pockets.

SERF1a is a strongly positively and negatively charged protein interacting with negatively charged RNA or mostly negatively charged α -synuclein. The distribution of charged residues along the hC helix structures detected in the CYANA conformations may play a role in this interaction. The hC helix (Figure 9) interestingly displays, on one face, a large patch of polar residues (green surface). This concentration of polar residues may increase the helix stability due to cooperative interaction between the polarizable side chains of such

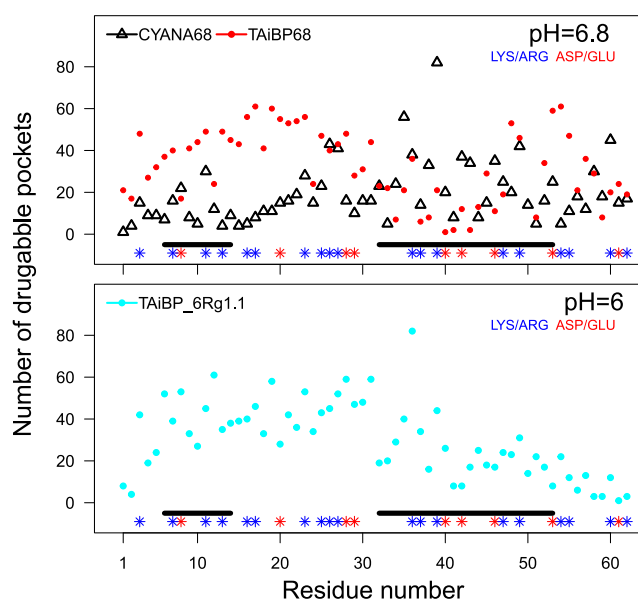


Figure 8. Number of druggable pockets predicted by fpocket⁴⁸ and including a given protein residue, plotted along the SERF1a primary sequence. The pockets were calculated using α -sphere radii in the 3–6 Å range and were selected as druggable if the druggability score was larger than 0.5. The plots are drawn for the set of filtered conformations CYANA68 (Δ), TAiBP68 (red bullets), and TAiBP6_Rg1.1 (cyan bullets). The positions of basic and acidic residues along the primary sequence are marked with blue and red asterisks. The positions of the α -helix consensus regions hN and hC are indicated with thick black lines.

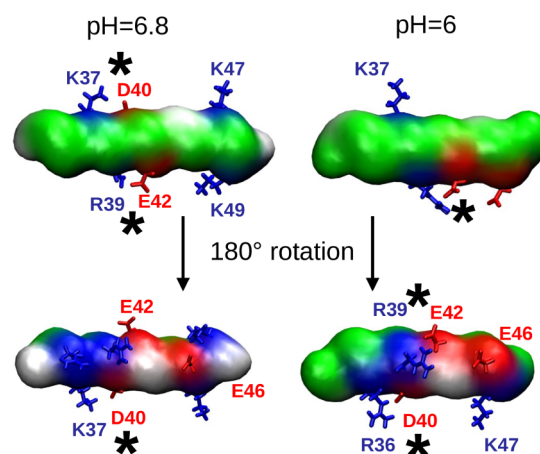


Figure 9. Front and rear views of α hC observed in CYANA conformations calculated at pH 6 and 6.8 are shown, with the backbone drawn with surface and colored according to the type of amino acids (acidic residues: red, basic residues: blue, polar residues: green, hydrophobic residues: white). The side chains of residues LYS, ARG, and ASP are drawn in licorice. The couple of basic-acidic residues with interacting side chains are indicated with asterisks.

residues. Among the acidic and basic residues, shown in licorice in Figure 9, several residues form couples of basic–acidic residues with interacting side chains, such as R39–E42, K37–D40, and R36–D40. The formation of these salt bridges has a stabilizing effect on the structure of the α helix and can also have a screening effect during the establishment of interactions with charged partners within a condensate. This

effect could be in line with the screening effect induced by the presence of counterions in the case of coacervation.⁴⁹

DISCUSSION

The present work focused on the analysis of the conformational space and internal dynamics of the Small EDRK-Rich Factor 1A (SERF1a) protein, which is a flexible protein known to play an important role in the fibrillation of α -synuclein. The conformational space of SERF1a has been studied in the past,^{6,8} at acidic or neutral pH values (6 or 6.8), but here we present the first comparative study of protein behavior at these two pH values.

The conformational space of the SERF1a protein was explored using two complementary approaches: (i) obtaining sets of conformations using the classical CYANA approach to determine the NMR structure⁵⁰ and (ii) enumerating conformations verifying the bound limits in the regions of ϕ and ψ values using the Threading-Augmented interval Branch-and-Prune (TAiBP).^{20,51} In particular, a method was developed to transform δ 2D prediction²³ of secondary structures from chemical shifts into ϕ , ψ boxes suitable for TAiBP input. The added value of the TAiBP approach, compared with existing ensemble-generation strategies such as ENSEMBLE¹⁹ and Flexible-Meccano,^{16,17} is that it perform an enumeration of conformations, allowing a systematic parsing of the Ramachandran diagram. The proposed pipeline, based on statistics of Ramachandran regions on triplets of amino acids, allows us to take into account the relative populations of loop regions of the Ramachandran diagram that are more present in disordered proteins. All NMR conformations generated by using CYANA or TAiBP were then filtered by fitting SEC-SAXS curves to propose representative sets of conformations.

Looking at the literature, SERF1a has been qualified as an intrinsically disordered protein (IDP) or a partially folded protein. Indeed, the observation of stable α -helices in this protein^{5,8} proves a strong level of residual folded structure. SERF1a is therefore an intermediate protein between intrinsically disordered proteins and flexible folded ones. More generally, the presence of α -helices surrounded by unfolded regions gives SERF1a a relationship to the structures of ribosomal proteins determined in the ribosome context. In particular, extended parts of ribosomal proteins have been found to be adapted to a specific environment.^{52,53} Similarly, adaptation of the unfolded regions of SERF1a to numerous RNA conformations could be essential for RNA-chaperon activity.

The use of CYANA for the calculation of the conformations reflects the ambiguity of SERF1a between an IDP and a partially folded protein. In fact, it is rather positive that the CYANA conformations calculated at pH 6.8 are able to produce the same quality of SEC-SAXS curve fit as the TAiBP68 conformations. At pH 6, the use of CYANA has permitted the detection of the long-range NOEs and, consequently, a finer interpretation of the experimental SEC-SAXS curves, applying a slight bias to the TAiBP6 conformations to increase the radius of gyration and produce a good fit of the SEC-SAXS curve.

Overall, the sets of TAiBP conformations are in good agreement with the experimental observations of the internal dynamics. In fact, the variability of the α -helix conformations at both pH values is correlated with the internal mobility observed by NMR relaxation. Furthermore, the profiles of S^2 values calculated from the TAiBP conformations or predicted

from the SERF1a primary sequence agree with the profiles of experimental S^2 values. For all these reasons, the use of TAiBP adds additional value to the description of the conformational space of SERF1a by allowing for a more refined characterization of the relationships between the conformational space and the internal dynamics of SERF1a.

The titration of SERF1a by α -synuclein at acidic pH reveals that a molar ratio of 1:5 of SERF1a: α -synuclein is necessary to detect chemical shift perturbations in the SERF1a region 13–28. In agreement with this observation, the reduction of the length of the α -helix in the hN region at acidic pH induces the appearance of numerous druggable pockets in the loop connecting the hN and hC regions, making it more prone to interaction with α -synuclein. Although the ITC measurements showed the specificity of the interaction between SERF1a and α -synuclein, it should be noted that the detection of druggable pockets by computational approaches does not demonstrate the specificity of pocket residues in the interactions between SERF1a and α -synuclein. Such a demonstration could be based on the mutagenesis of residues most involved in the pocket definition and on the analysis of the interaction of mutated SERF1a with α -synuclein.

From a more general perspective, searching for binding pockets in the SERF1a conformations paves the way toward the discovery of compounds interfering with the pathological effect of this protein. In this way, the present study follows in the footsteps of previous studies.^{54–56} The druggable pockets detected in the present work can be considered potential cryptic pockets⁵⁷ as they appear only in some of the SERF1a conformations.

The relatively large molar ratio of SERF1a: α -synuclein, for which chemical shift perturbations are observed, corresponds to situations of α -synuclein excess with respect to SERF1a. This situation might be compatible with the interaction of several extended α -synucleins with one crossing SERF1a, which favors the formation of a β -sheet between α -synuclein monomers, the precursor of fibrils. Indeed, the least stable N-terminal α -helix at pH 6, in addition to inducing the appearance of a larger number of druggable pockets in the region of 13–28 residues, favors extended conformations of SERF1a in this region. These relatively extended conformations are more prone to interacting with several extended chains of α -synuclein. In that perspective, the hamster SERF1a was shown¹¹ to display extended conformations during its interaction with α -synuclein. The quantity of α -synuclein, which is much larger than the quantity of SERF1a, also agrees with the catalytic role of SERF1a in the formation of fibrils.

The effect of acidic pH on SERF1a conformations agrees with the cellular context of SERF1a function. In fact, acidic pH is observed in the nucleolus ultrastructure, in which SERF1a operates its physiological function. Furthermore, in the case of some amyloid proteins, acidification was shown to have an effect on phase separation.^{58,59}

In liquid–liquid phase separation (LLPS), such as the phase corresponding to the nucleolus,⁶⁰ or the phase SERF1a was shown to induce with α -synuclein¹¹ or with TAR RNA,⁴ the balance between electrostatic repulsion of the polymers and solvation of these polymers in the condensate plays an essential role in the formation of this phase.⁶¹ The analysis of hC helix conformations in SERF1a permits us to propose a model based on the role of acidic-basic pairs of residues to screen electrostatic repulsion.

In summary, the present study of SERF1a using solution-state NMR and SEC-SAXS gives insight into the variations of conformations at two pH values. The simultaneous use of a purely geometric approach, along with classical methods for NMR structure determination, provides a more in-depth description of the conformational space, which is filtered in a complementary way by the corresponding SEC-SAXS data. In that framework, the behavior of SERF1a during its interactions with molecular partners can be extrapolated, providing a foundation for clarifying the molecular aspects of the physiological and pathological actions of SERF1a. In particular, the detection of druggable pockets in the sets of SERF1a conformations is a starting point toward the building of interaction models between SERF1a and α -synuclein and thus presents a drug discovery potential.

MATERIALS AND METHODS

Purification and Preparation of Samples

SERF1a (NP_001171558.1, 62 a.a.) was expressed by *E. coli* BL21(DE3) system as a fusion protein with the maltose-binding protein (MBP) and six histidines. The NMR sample was prepared by culture in M9 minimal medium with 4g/liter $^{15}\text{NH}_4\text{Cl}$ ammonium chloride and 2g/liter $^{13}\text{C}_6\text{H}_{12}\text{O}_6$ D-glucose. The fusion protein was isolated by immobilized metal affinity chromatography (IMAC) resin (Cytiva) in 25 mM Tris buffer (pH 8), 300 mM NaCl, 10 mM 2-mercaptoethanol, 10 mM imidazole for washing, and 300 mM imidazole for elution. The elution was dialyzed in 25 mM Tris (pH 8), 100 mM NaCl, and 50 mM 2-mercaptoethanol buffer and then cleaved by Tobacco Etch Virus (TEV) protease at 30 °C for 3 h. To remove MBP, His tag, and other impurities, the protein solution was passed through a cation exchange column (TSKgel) in 20 mM NaPi (pH 7.7). SERF1a was eluted at approximately 195 mM NaCl by a linear gradient under 20 mM NaPi (pH 7.7) and 1 M NaCl. The collected fractions were concentrated and further purified by a size-exclusion column, Superdex 75 Increase 10/300 (Cytiva) in 20 mM NaPi (pH 6.0 or pH 6.8), 20 mM NaCl, and 0.02% NaN_3 . The eluted fractions were confirmed by SDS-PAGE and then concentrated by a 1 kDa centrifugal filter (Pall). Size-exclusion chromatography-based SAXS (SEC-SAXS) data were measured using the TPS 13A beamline of NSRRC, Hsinchu.⁶²

DISCLOSURE

The NMR chemical shifts measured at pH 6.8 and 6 were deposited to the BMRB (BMRB IDs 36733 and 36732). The initial conformations generated using CYANA from NMR measurements at pH 6.8 and 6 were deposited to the Protein Data Bank (PDB IDs 9M2D and 9M27). The CYANA and TAI BP conformations filtered using the SEC-SAXS data are provided as supporting material of the manuscript.

The SEC-SAXS data were deposited in the Small-Angle Scattering Biological Data Bank (SASDWZ9 and SASDWY9).

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsomega.5c07620>.

Conformation of filtered conformations (ZIP)

Additional experimental details and materials and methods(PDF)

AUTHOR INFORMATION

Corresponding Authors

Jung-Hsin Lin – Biomedical Translation Research Centre, Academia Sinica, Taipei 115201, Taiwan; Research Center for Applied Sciences, Academia Sinica, Taipei 11529, Taiwan; School of Pharmacy, College of Medicine, National Taiwan University, Taipei 100025, Taiwan; College of Engineering Sciences, Chang Gung University, Taoyuan 33302, Taiwan; orcid.org/0000-0002-6712-4994; Email: jhlin@gate.sinica.edu.tw

Thérèse E. Malliavin – Université de Lorraine, CNRS, LPCT, Nancy F-54000, France; Université de Lorraine, CNRS, LPCT, Metz F-57000, France; orcid.org/0000-0002-3276-3366; Email: therese.malliavin@univ-lorraine.fr

Authors

Shu-Yu Huang – Research Center for Applied Sciences, Academia Sinica, Taipei 11529, Taiwan; orcid.org/0009-0002-0826-2454

Orion Shih – National Synchrotron Radiation Research Center (NSRRC), Hsinchu City 300092, Taiwan; orcid.org/0000-0002-0610-6856

U-Ser Jeng – National Synchrotron Radiation Research Center (NSRRC), Hsinchu City 300092, Taiwan; Department of Chemical Engineering & College of Semiconductor Research, National Tsing Hua University, Hsinchu 300044, Taiwan; orcid.org/0000-0002-2247-5061

Chi-Fon Chang – Genomics Research Center, Academia Sinica, Taipei 11529, Taiwan; orcid.org/0000-0002-2961-0748

Complete contact information is available at:

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Author Contributions

S.-Y.H. performed the NMR measurements and the CYANA structure calculations, O.S. performed the SEC-SAXS measurements, U.-S.J. conceptualized the work and participated in the manuscript writing, C.-F.C. participated in the NMR analysis and in the manuscript writing, J.-H.L. conceptualized the work and wrote the manuscript, T.E.M. realized the TAI BP structure calculations and wrote the manuscript. All authors participated in the discussions and the revision of the manuscript.

Notes

The authors declare no competing financial interest.

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